

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.3: Compression

Concept 1.3.3: Compression Methods (Run-Length Encoding)

Total Questions: 7

Mark Schemes begin on Page 5

Question 1 | Source: 9618_w25_qp_11 (Q7.d.iii)

7 Web browsers use Internet Protocol (IP) addresses.

(d) A bitmap file is downloaded. The image has a maximum of 256 colours and measures 512 pixels wide by 2048 pixels high.

(iii) Explain how a bitmap image is compressed using run-length encoding (RLE).

.....

.....

.....

..... [2]

Question 2 | Source: 9618_w24_qp_11 (Q4.b.ii)

4 A student uses a laptop to write a program that is saved as a text file.

(b) The student compresses the file before it is emailed to their teacher as an attachment.

(ii) Describe **one** lossless method of compressing a text file.

.....

.....

.....

.....

.....

..... [3]

Question 3 | Source: 9618_w24_qp_13 (Q6.c.ii)

6 A computer system stores text, images and sound.

(ii) Describe **one** method of compressing a sound file using lossy compression.

.....

.....

.....

..... [2]

Question 4 | Source: 9618_s24_qp_13 (Q2.b.ii)

2 A photograph is stored as a bitmap image.

(b) The photograph is compressed before being uploaded to a web server.

(ii) Explain how run-length encoding (RLE) will compress the photograph.

.....

.....

.....

..... [2]

Question 5 | Source: 9618_s23_qp_11 (Q1.c.ii)

1 Images are being created to advertise holidays.

Some of the images are bitmap images and some are vector graphics.

(c) The photographs are compressed before they are uploaded to a web server.
Customers download the photographs from this web server.

(ii) An image can be compressed using run-length encoding (RLE).

Explain the reasons why RLE may **not** reduce the file size of a bitmap image.
Give **one** example in your answer.

.....

.....

.....

.....

.....

..... [3]

Question 6 | Source: 9618_w22_qp_11 (Q7.c)

(c) An Operating System may include a utility program to compress text files.

Describe **one** appropriate method of compressing a text file.

.....

.....

.....

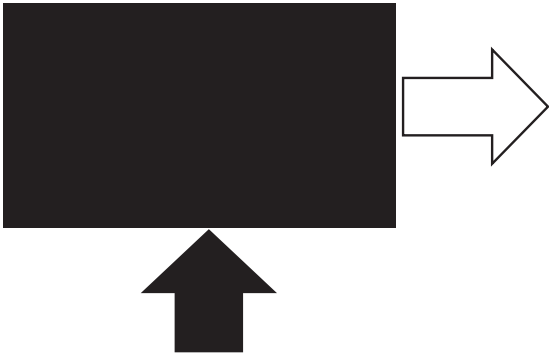
.....

.....

..... [3]

Question 7 | Source: 9618_w22_qp_12 (Q8.c.i)

- 8 The following bitmap image has a resolution of 4096 × 4096 pixels and a colour depth of 24 bits per pixel.



The image is displayed on a monitor that has a screen resolution of 1920 × 1080 pixels.

- (c) A second bitmap image is stored using a colour depth of 8 bits per pixel.

The file is compressed using run-length encoding (RLE).

- (i) The table shows the compressed and uncompressed values for parts of the image file.

Each colour of the pixel in the image is represented by a hexadecimal value.

Complete the table. The first row has been completed for you.

Uncompressed image	RLE compressed image
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA
	2AB 2FF 11D 167
32 32 80 81 81	

[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q7.d.iii)

7(d)(iii)	1 mark per bullet point, max 2 marks - Sequences of consecutive identical colours / pixels ... are stored as the colour value and the number of times it occurs consecutively	2
-----------	--	----------

Mark Scheme for Question 2 | Source: 9618_w24_ms_11 (Q4.b.ii)

4(b)(ii)	1 mark for each bullet point (max 3) - Run-Length Encoding // RLE - Repeated sequences of the same characters are replaced by ... a single copy of the character ... and a count of the number of characters	3
----------	---	----------

Mark Scheme for Question 3 | Source: 9618_w24_ms_13 (Q6.c.ii)

6(c)(ii)	1 mark for each correct point and 1 mark for matching expansion e.g. - Decrease sample rate ... fewer samples/readings/measurements stored per second // fewer bits per second stored - Decrease sample resolution ... fewer bits per sample/reading/measurement // each sample has fewer bits - Sound outside of set/human hearing range is removed ... fewer measurements are stored / decreases the number of possible binary values so fewer bits are stored	2
----------	--	----------

Mark Scheme for Question 4 | Source: 9618_s24_ms_13 (Q2.b.ii)

2(b)(ii)	1 mark each: - Identifies consecutive repeating pixels of the same colour - Stores the colour /pattern and the number of times it repeats	2
----------	---	----------

Mark Scheme for Question 5 | Source: 9618_s23_ms_11 (Q1.c.ii)

1(c)(ii)	1 mark each to max 2 for explanation; 1 mark for an image related example - RLE stores a colour and the number of times it occurs consecutively - An image may not have many sequences of the same colour - It would need to store each colour and then the count/number 1 which adds data Example: - Red-Green-Blue would become Red 1 Green 1 Blue 1	3
----------	---	----------

Mark Scheme for Question 6 | Source: 9618_w22_ms_11 (Q7.c)

7(c)	1 mark for each bullet point (max 3): - lossless compression - Run Length Encoding - repeated sequences of <u>characters</u> are replaced by ... a single copy of the character ... and a counter of the number of characters	3
------	---	----------

Mark Scheme for Question 7 | Source: 9618_w22_ms_12 (Q8.c.i)

8(c)(i)

1 mark for each correct underlined part:

2

Uncompressed sound	RLE compressed sound
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA
<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167
32 32 80 81 81	<u>232 180 281</u>